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produced. Phosphorylation is the process of synthesis of ATP by the addition of inorganic phosphate to ADP. The addition of phosphate here takes place with the help of light generated electron and so it is called as **photophosphorylation**. It takes place in both cyclic and non-cyclic electron transport.

13.11.1 Cyclic Photophosphorylation

Cyclic photophosphorylation refers to the electrons ejected from the pigment system I (Photosystem I) and again cycled back to the PS I. When the photons activate P700 reaction centre photosystem II is activated. Electrons are raised to the high energy level. The primary electron acceptor is Ferredoxin Reducing Substance (FRS) which transfers electrons to Ferredoxin (Fd), Plastoquinone (PQ), cytochrome b6-f complex, Plastocyanin (PC) and finally back to chlorophyll P700 (PS I). During this movement of electrons Adenosine Di Phosphate (ADP) is phosphorylated, by the addition of inorganic phosphate and generates Adenosine Tri Phosphate (ATP). Cyclic electron transport produces only ATP and there is no NADPH + H^+ formation. At each step of electron transport, electron loses potential energy and is used by the transport chain to pump H^+ ions across the thylakoid membrane. The proton gradient triggers ATP formation in ATP synthase enzyme situated on the thylakoid membrane. Photosystem I need light of longer wave length (> P700 nm). It operates under low light intensity, less CO_2 and under anaerobic conditions which makes it considered as earlier in evolution (Figure 13.15).

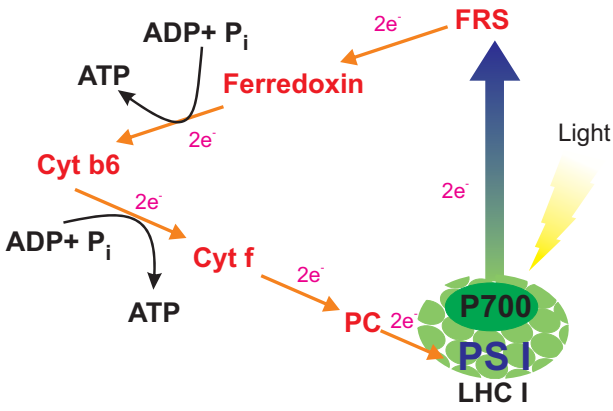


Figure 13.15: Cyclic Photophosphorylation

13.11.2 Non-Cyclic Photophosphorylation

When photons are activated reaction centre of pigment system II(P680), electrons are moved to the high energy level. Electrons from high energy state passes through series of electron carriers like pheophytin, plastoquinone, cytochrome complex, plastocyanin and finally accepted by PS I (P700). During this movement of electrons from PS II to PS I ATP is generated (Figure 13. 16). PS I (P700) is activated by light, electrons are moved to high energy state and accepted by electron acceptor molecule ferredoxin reducing Substance (FRS). During the downhill movement through ferredoxin, electrons are transferred to $NADP^+$ and reduced into NADPH + H^+ (H^+ formed from splitting of water by light).

Electrons released from the photosystem II are not cycled back. It is used for the reduction of $NADP^+$ in to NADPH + H^+. During the electron transport it generates ATP and hence this type of photophosphorylation is called **non-cyclic photophosphorylation**. The electron flow looks like the appearance of letter 'Z' and so known as **Z scheme**. When there is availability of $NADP^+$ for reduction and when there is splitting of